Lista de Trigonometria

1 Sejam $\frac{\pi}{18} = a$ e $\frac{\pi}{9} = b$. Reescrevendo: $\frac{tga + tgb}{1 - tga\, tgb} = tg(a+b) =$

$= tg\left(\frac{\pi}{18} + \frac{\pi}{9}\right) = tg\frac{\pi}{6} = \frac{\sqrt{3}}{3}$

2. $tg(a-b) = \sqrt{3} \Rightarrow \frac{\overbrace{tga}^{1} - \overbrace{tgb}^{x}}{1 + \underbrace{tga}_{1}\,\underbrace{tgb}_{x}} = \sqrt{3} \Rightarrow \frac{1-x}{1+x} = \sqrt{3} \Rightarrow \sqrt{3} + x\sqrt{3} = 1 - x \Rightarrow$

$\Rightarrow x\sqrt{3} + x = 1 - \sqrt{3} \Rightarrow x(\sqrt{3}+1) = 1 - \sqrt{3} \Rightarrow x = \frac{1-\sqrt{3}}{1+\sqrt{3}} \cdot \frac{(1-\sqrt{3})}{(1-\sqrt{3})} = \frac{(1-\sqrt{3})^2}{-2} =$

$= \frac{1 - 2\sqrt{3} + 3}{-2} = \frac{4 - 2\sqrt{3}}{-2} = -2 + \sqrt{3}$ $\quad \therefore tg\, b = -2 + \sqrt{3}$

3. ΔABD: $tg\beta = \frac{2}{3}$

ΔABC: $tg(\alpha+\beta) = \frac{t}{3} \Rightarrow \frac{\overbrace{tg\alpha}^{x} + \overbrace{tg\beta}^{\frac{2}{3}}}{1 - \underbrace{tg\alpha}_{x}\,\underbrace{tg\beta}_{\frac{2}{3}}} = \frac{t}{3} \Rightarrow \frac{x + \frac{2}{3}}{1 - \frac{2}{3}x} = \frac{t}{3} \Rightarrow$

$\Rightarrow \frac{\frac{3x+2}{3}}{\frac{3-2x}{3}} = \frac{t}{3} \Rightarrow 9x + 6 = 21 - 14x \Rightarrow 23x = 15 \Rightarrow x = \frac{15}{23}$ $\quad \therefore tg\,\alpha = \frac{15}{23}$

4. Seja $\frac{\pi}{8} = a \Rightarrow \cos^2 a - \operatorname{sen}^2 a = \cos 2a$. Assim,

$\cos^2\frac{\pi}{8} - \operatorname{sen}^2\frac{\pi}{8} = \cos\left(2 \cdot \frac{\pi}{8}\right) = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$

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5. Seja $\frac{\pi}{24} = a \Rightarrow (\cos a - \sin a)^2 = \cos^2 a - \underbrace{2\cos a \cdot \sin a}_{\sin 2a} + \sin^2 a = \overbrace{1}^{1} - \sin 2a$

$= 1 - \sin\left(2 \cdot \frac{\pi}{24}\right) = 1 - \sin\frac{\pi}{12} = 1 - \sin 15° \quad (*)$

$\sin 15° = \sin(45° - 30°) = \underbrace{\sin 45°}_{\frac{\sqrt{2}}{2}} \cdot \underbrace{\cos 30°}_{\frac{\sqrt{3}}{2}} - \underbrace{\sin 30°}_{\frac{1}{2}} \cdot \underbrace{\cos 45°}_{\frac{\sqrt{2}}{2}} = \frac{\sqrt{6} - \sqrt{2}}{4}$

Substituindo em $(*)$: $1 - \left(\frac{\sqrt{6} - \sqrt{2}}{4}\right) = \frac{4 - \sqrt{6} + \sqrt{2}}{4}$

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6. Seja $a = 35°$. Assim, $\sin 2a = 0{,}94$

$A = \sin a \cdot \cos a = ?$ $\quad \Rightarrow \quad 2\sin a \cdot \cos a = 0{,}94 \Rightarrow \sin a \cos a = 0{,}47$

$\therefore A = 0{,}47$

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7. $\operatorname{tg} x + \operatorname{cotg} x = 3 \Rightarrow \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = 3 \Rightarrow \frac{\overbrace{\sin^2 x + \cos^2 x}^{1}}{\cos x \sin x} = 3 \Rightarrow \cos x \cdot \sin x = \frac{1}{3}$

a) $\sin 2x = 2 \cdot \underbrace{\sin x \cdot \cos x}_{1/3} = \frac{2}{3}$

b) $\sin^2(2x) + \cos^2(2x) = 1 \Rightarrow \frac{4}{9} + \cos^2(2x) = 1 \Rightarrow \cos^2(2x) = \frac{5}{9} \Rightarrow \cos 2x = \pm\frac{\sqrt{5}}{3}$

↓ Relação fundamental!

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8. 5. $\underbrace{\cos(2x)}_{-2\sin^2 x + 1} + 3\sin x = 4$ $\quad \sin x$ e $\cos x$?

$5.(-2\sin^2 x + 1) + 3\sin x = 4 \Rightarrow -10\sin^2 x + 3\sin x + 1 = 0 \cdot (-1) \Rightarrow$

$\Rightarrow 10\sin^2 x - 3\sin x - 1 = 0$. Seja $\sin x = y$: $10y^2 - 3y - 1 = 0 \Rightarrow$

$\Rightarrow \underbrace{y}_{\sin x} = \dfrac{3 \pm \sqrt{9+40}}{20}$ $\begin{cases} \frac{3+7}{20} = \frac{1}{2} \\ \frac{3-7}{20} = -\frac{1}{5} \end{cases}$

Como $x \in 3^{\circ}Q$, $\sin x < 0$ e, portanto, $\sin x = -\frac{1}{5}$

$\underbrace{\sin^2 x}_{\frac{1}{25}} + \cos^2 x = 1 \Rightarrow \cos^2 x = 1 - \frac{1}{25} \Rightarrow \cos^2 x = \frac{24}{25} \Rightarrow \cos x = \pm \frac{2\sqrt{6}}{5}$

Como $x \in 3^{\circ}Q$, $\cos x < 0$ e, portanto, $\cos x = -\frac{2\sqrt{6}}{5}$

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9. Seja $\frac{x}{2} = a \Rightarrow x = 2a$. Reescrevendo:

$\frac{\pi}{2} \overset{:2}{<} 2a \overset{:2}{<} \pi \overset{:2}{}$; $\cos 2a = -\frac{7}{25}$; $\cos a = ?$

$\frac{\pi}{4} < a < \frac{\pi}{2}$ (*) fórmula $\quad 2\cos^2 a - 1 = -\frac{7}{25} \Rightarrow 2\cos^2 a = \frac{18}{25} \Rightarrow \cos a = \pm \frac{3}{5}$

De (*), $a \in 1^{\circ}Q \therefore \cos a = \cos \frac{x}{2} = \frac{3}{5}$

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